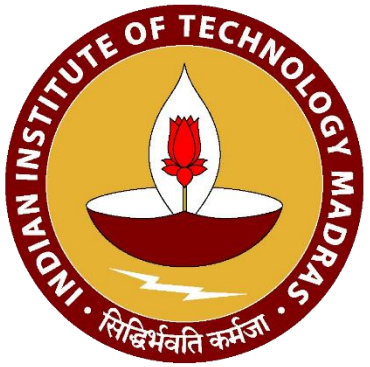
DA5400: Foundations of Machine Learning

Topic 4: Regression – Part 2



Outline

- Gradient Descent Approach to Linear Regression
- Supervised Learning Workflow
- Bias Vs Variance in Regression Models
- Regularisation in Regression Models

Recap: OLS Solution to Linear Regression

- Model Structure: $\hat{y} = w_0 + w_1 x_1 + \cdots + w_D x_D$
- Known: Mapped Input $(\boldsymbol{X})$ and Output Data $(\boldsymbol{y})$
- Sum of squared residuals in predicting y is given by:

$$SSR = (\boldsymbol{y} - \boldsymbol{X}^* \boldsymbol{w})^T (\boldsymbol{y} - \boldsymbol{X}^* \boldsymbol{w})$$

- Least squares estimate of $\boldsymbol{w}$ are the coefficients that minimize SSR

$$\hat{\boldsymbol{w}} = \underset{\boldsymbol{w}}{\operatorname{argmin}} SSR$$

- Result: Analytical solution to the least squares estimate of $\boldsymbol{w}$ is given by:

$$\hat{\boldsymbol{w}} = \left(\boldsymbol{X}^{*T} \boldsymbol{X}^*\right)^{-1} \boldsymbol{X}^{*T} \boldsymbol{y}$$ – Normal Equation

Limitations of Closed Form Solution

- Requires computing $\left(X^{*T}X^{*}\right)^{-1}$
- Computationally expensive for large datasets
- Requires storing all data in memory
- Cannot be used when data is streaming in batches

- Q: Do we really need to find the matrix inverse?

- Not really. OLS is fundamentally an optimization problem and can be solved iteratively

Gradient Descent for Linear Regression

Loss Function and Gradient Descent

- Loss Function for Simple Linear Regression:

$$J(w) = \frac{1}{2N}(\boldsymbol{y} - \boldsymbol{X}^*\boldsymbol{w})^T(\boldsymbol{y} - \boldsymbol{X}^*\boldsymbol{w})$$

$$J(w) = \frac{1}{2N}\sum_{i=1}^{N}\left(y_i - \boldsymbol{x}_i^T\boldsymbol{w}\right)^2$$

- Properties of Loss Function:
  - Quadratic and Convex
  - Global minimum exists
  - Ideal for gradient based optimization

Gradient Descent Approach

- Principle of Gradient Descent:
  - Start with an initial guess of parameters (w)
  - Iteratively move in the direction of steepest descent of the cost function

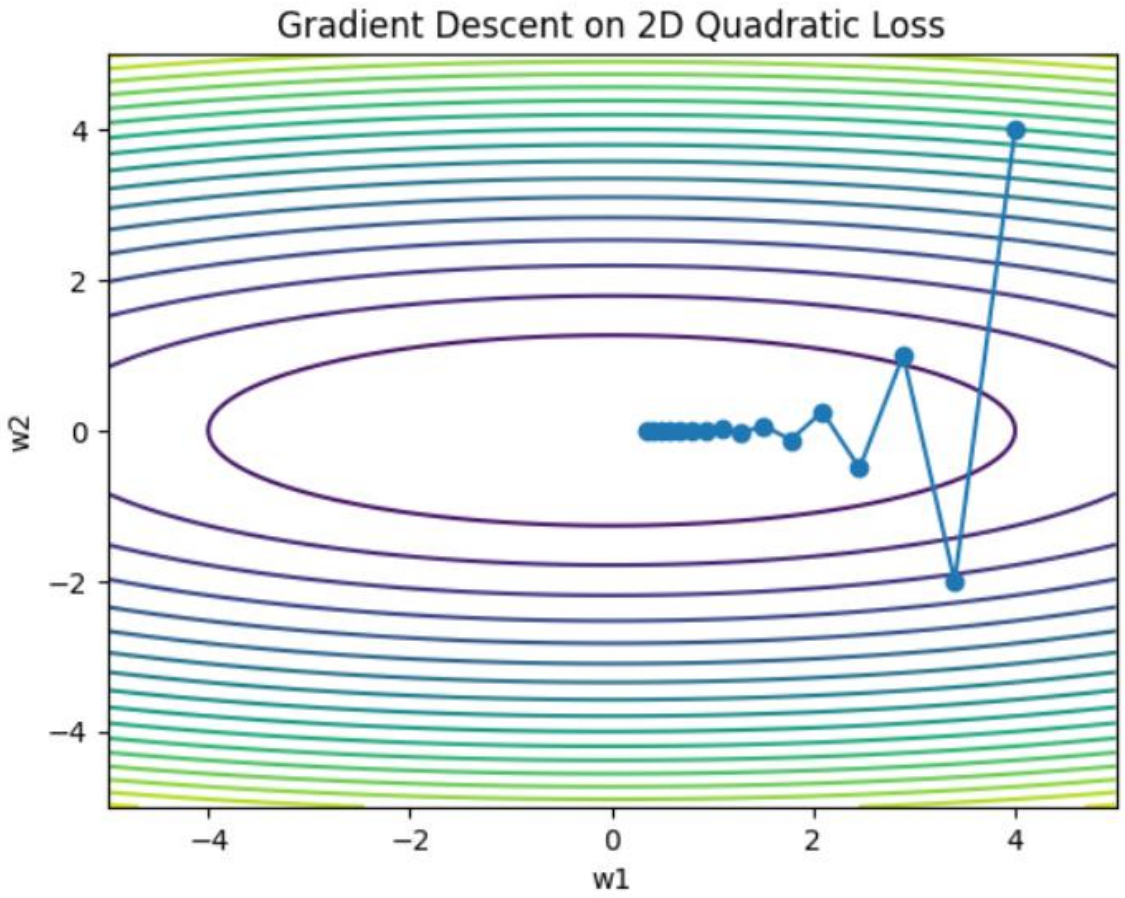
$$\boldsymbol{w}^{k+1} = \boldsymbol{w}^{k} - \alpha \nabla J(\boldsymbol{w}^{k})$$

- For Linear Regression:

$$\nabla J(\boldsymbol{w}) = -\frac{1}{N}\boldsymbol{X}^{T}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w})$$

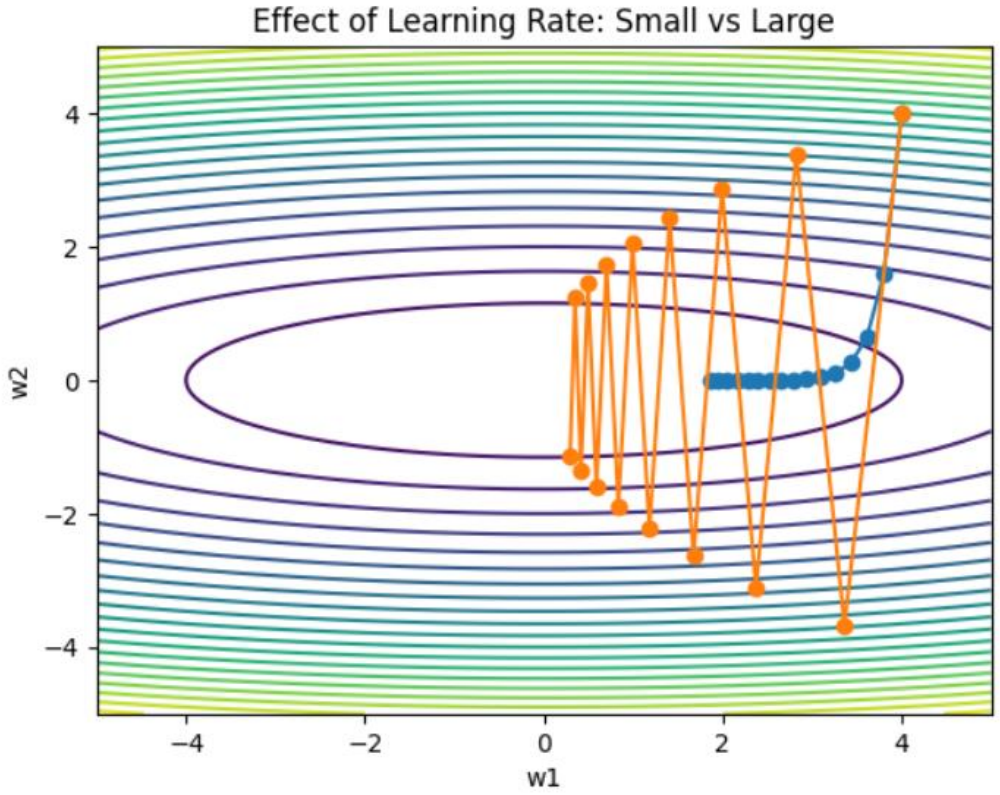
- Gradient update rule:

$$\boldsymbol{w}^{k+1} = \boldsymbol{w}^{k} + \alpha \frac{1}{N}\boldsymbol{X}^{T}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w})$$



Learning Rate α

- Gradient gives only the direction of steepest descent, not how far to move
- α controls the step size along the gradient
- Choice of alpha:
  - Should be balanced and not too small or too large
  - Too small – Very slow learning
  - Too large – Can jump over the minimum or oscillate and diverge

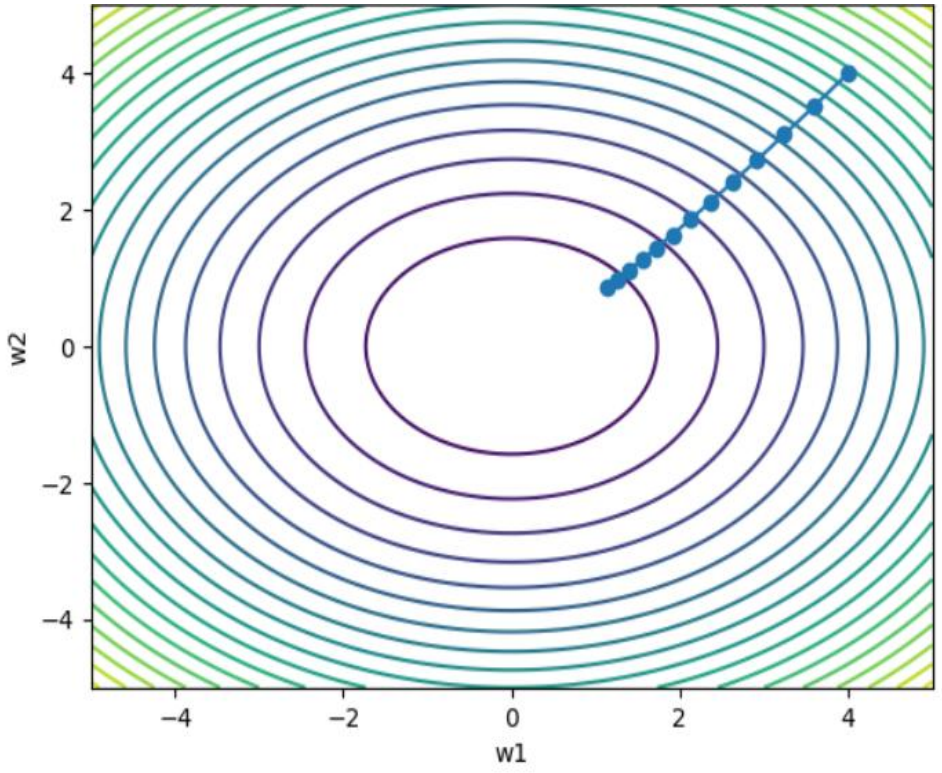


Feature Scaling

- Features are recommended to be scaled to standard normal in gradient descent approach

$$x' = \frac{x - \bar{x}}{\sigma_x}$$

- Without scaling, features with large magnitude dominate the gradient
- Geometrically without scaling:
  - Contours are more elliptical
  - Learning rate should be lower for stability
  - Gradient descent would be slow to converge
- With scaling:
  - Contours are more circular
  - Learning rate can be higher as gradient closely aligns with direction to minimum
  - Smooth and short path to convergence



Gradient Descent and Normal Equation

- Gradient descent converges to the same solution as OLS when
  – learning rate is chosen properly
  – enough iterations are run
- Q: How can this be proven mathematically?

- At optimum:

$$\nabla J(\boldsymbol{w}) = 0$$

$$\frac{1}{N}\boldsymbol{X}^T(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}) = 0$$

$$\hat{\boldsymbol{w}} = \left(\boldsymbol{X}^{*T}\boldsymbol{X}^*\right)^{-1}\boldsymbol{X}^{*T}\boldsymbol{y}$$

Supervised Learning Workflow

Supervised Learning Workflow

Translating a Business Problem into SL Problem

- Identify the business objective
- Translate it into a prediction problem
- Define target and input variables
- Define success metrics

Data Collection & Understanding

- Gather input-output pairs
- Understand the data
  - Noise
  - Missing values
  - Data ranges and units
  - Correlations

Data Pre-processing

- Create informative predictors
- Feature Selection
- Feature Transformation

Supervised Learning Workflow

Hypothesis Selection

- Choose model family
  - Linear or Non-linear
  - Which variant of linear
  - Non-linear structure

Feature Engineering

- Create informative predictors
- Feature Selection
- Feature Transformation
- Feature Scaling

Model Training (Estimation)

- Define Loss function
- Estimate parameters
- Fine tune the hyperparameters

Supervised Learning Workflow

Model Validation

- Evaluate on data not used for training based on different metrics
- Detect underfitting and overfitting
- Repeat from hypothesis selection if needed

Model Selection & Testing

- Choose model with best validation performance
- Assess performance on a hold out set
- Provides an unbiased estimate of error

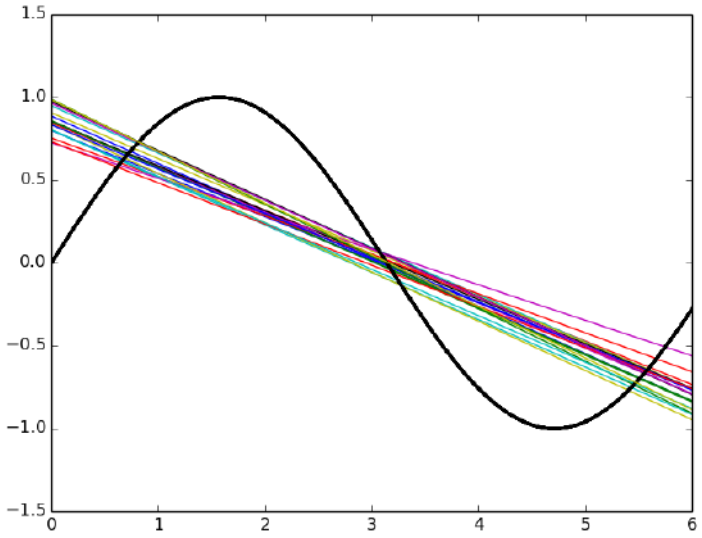
Model Deployment & Maintenance

- Deploy model for predictions
- Monitor the performance regularly
- Check for prediction drift, input distribution drift, need for re-training, etc.

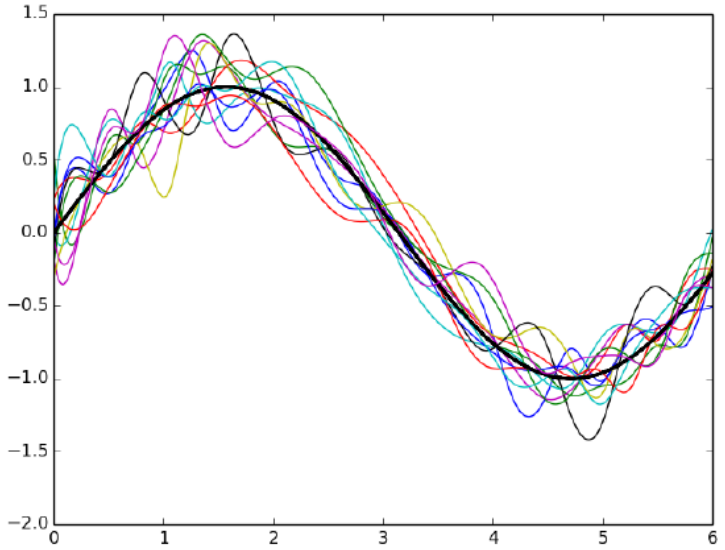
Bias Vs Variance in Regression Models

Bias Vs Variance: Motivation

- Consider a true function $y = \sin x$
- Suppose we draw multiple sample sets (say $N = 30$) with some noise and fit a linear model and high degree polynomial model on each of them



Simple model: Under-fitting

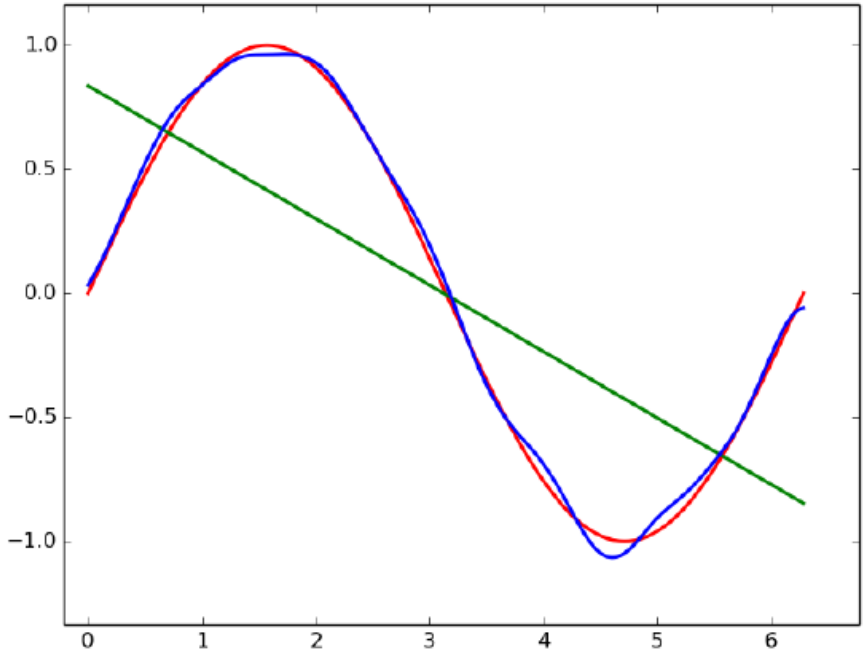


Complex model: Over-fitting

Bias in Predictions

- Bias in predictions measures how far average prediction is from true value

$$Bias = \mathbb{E}[\hat{y}] - y^t$$



Green Line: Average value of $\hat{y}$ for a linear model – High bias

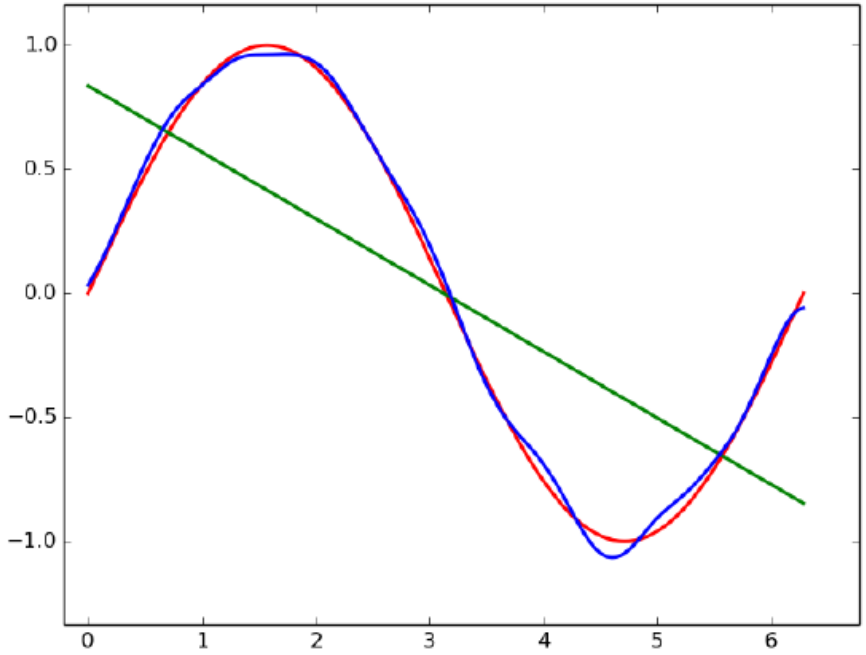
Blue Curve: Average value of $\hat{y}$ for a complex model – Low bias

Red Line: True model (y^t)

Variance in Predictions

- Variance measures change in model's predictions when trained on different datasets

$$Var = \mathbb{E}\left[\hat{y} - \mathbb{E}[\hat{y}]\right]^2$$



Simple model: High bias, Low Variance
Complex model: Low bias, High Variance
So bias Vs variance trade-off is inevitable in ML

Prediction Error due to Bias and Variance

- Prediction bias and variance have two distinct sources
- Model structure (Choice of hypothesis space):
  - Linear Vs Non-linear
  - Polynomial degree
  - Feature set
- Parameter Estimation:
  - Sample variability
  - Noise in observations
  - Multi-collinearity

- Note: Increasing data reduces variance but cannot remove model bias

Relation: Error, Bias and Variance

- Consider a new point (x, y) not used in training
- MSE in predicting the value of y using the trained model can be written as:

MSE = Bias^2 + Variance + Irreducible error

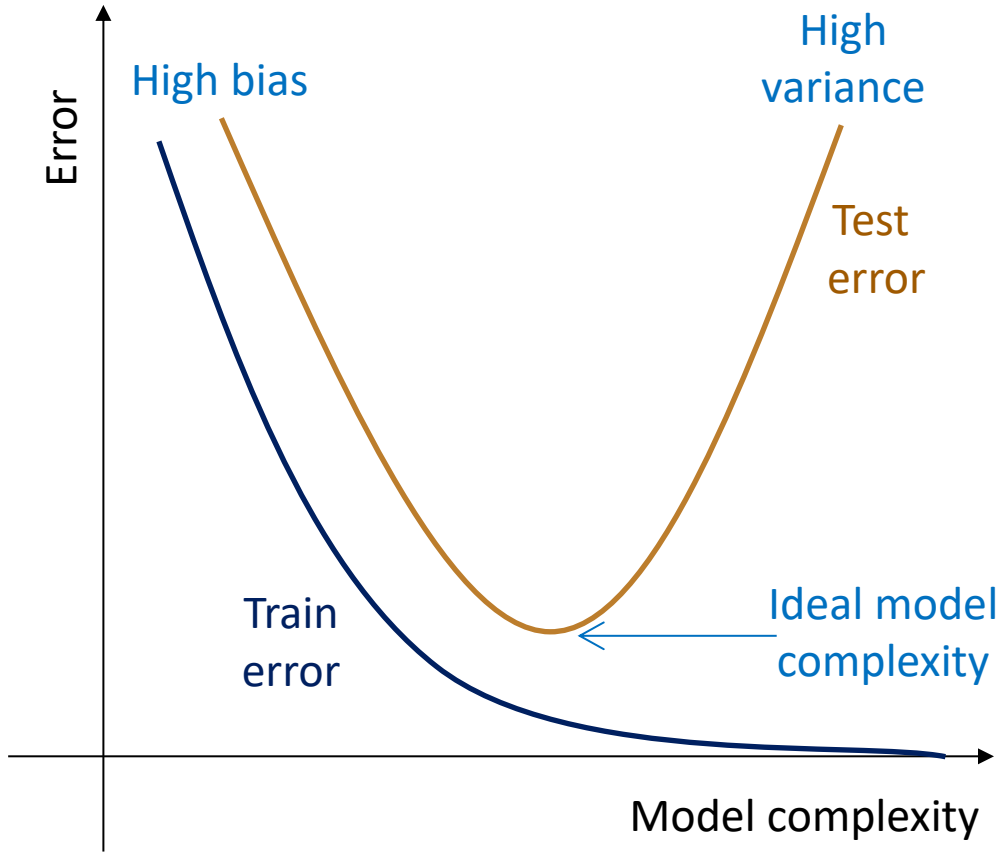
$$\mathbb{E}[(y - \hat{y})^2] = (\mathbb{E}[\hat{y}] - y^t)^2 + \mathbb{E}[(\hat{y} - \mathbb{E}[\hat{y}])^2] + \sigma_\epsilon^2$$

(Check here for proof)

Regularised Regression Models

Regularisation: Motivation

- Objectives of Regression:
  - Learn a model that explains training data and generalises to unseen data
  - Bias and variance should be balanced
  - Minimise test error and not just training error
- Idea:
  - Choose a model structure which allows some flexibility (high variance)
  - Regularise it to control that flexibility
  - Converts a high-variance model into a balanced model



Regularisation: Math Idea

- Regularised OLS objective:

$$\min_{\boldsymbol{w}} (\boldsymbol{y} - \boldsymbol{X}^* \boldsymbol{w})^T (\boldsymbol{y} - \boldsymbol{X}^* \boldsymbol{w}) + \lambda \Omega(\mathbf{w})$$

- $\Omega(\mathbf{w})$: Regularisation penalty – balances against training loss
- $\Omega(\mathbf{w})$ is large for complex models and small for simple models

- $\lambda \geq 0$: Regularisation rate – controls the strength of regularization

- Impact of regularization:
  - Parameters are pushed towards zero
  - Significant decrease in variance with small increase in bias

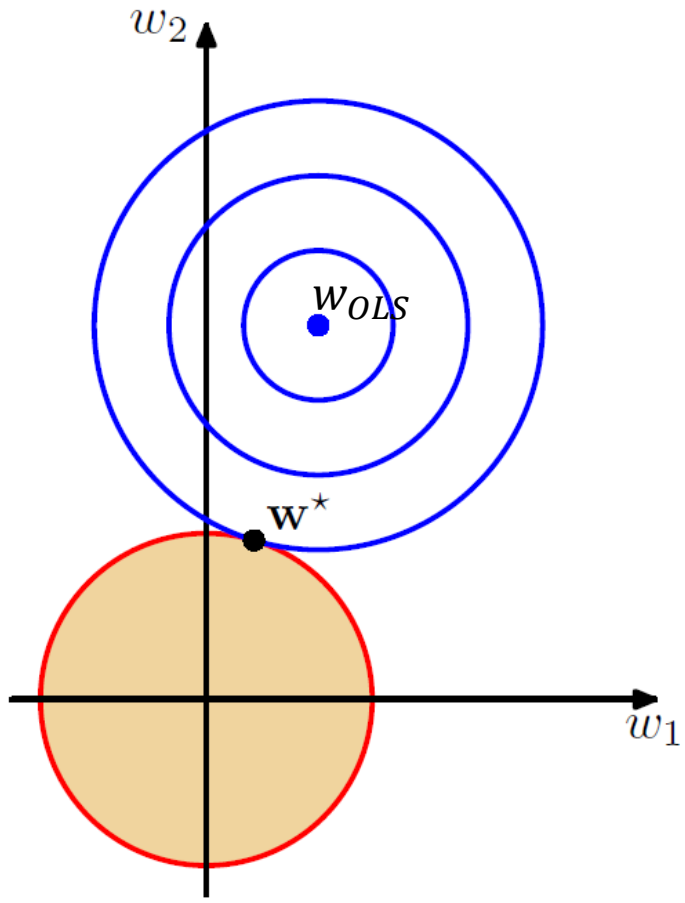
Ridge Regression (L2 Regularisation)

- Ridge regression objective:

$$\min_{\boldsymbol{w}} (\boldsymbol{y} - \boldsymbol{X}^{*}\boldsymbol{w})^{T}(\boldsymbol{y} - \boldsymbol{X}^{*}\boldsymbol{w}) + \lambda \left|\left|\mathbf{w}\right|\right|_{2}^{2}$$

$$\boldsymbol{w}_{ridge} = \left(\boldsymbol{X}^{*T}\boldsymbol{X}^{*} + \lambda \boldsymbol{I}\right)^{-1}\boldsymbol{X}^{*T}\boldsymbol{y}$$

- Penalises squared magnitude of coefficients
- Shrinks all coefficients smoothly with increase in λ
- Shrinks the larger coefficients more
- Geometric View:
  - OLS: Minimises error in an unconstrained space
  - Ridge: Minimises error constrained inside a L2 circle



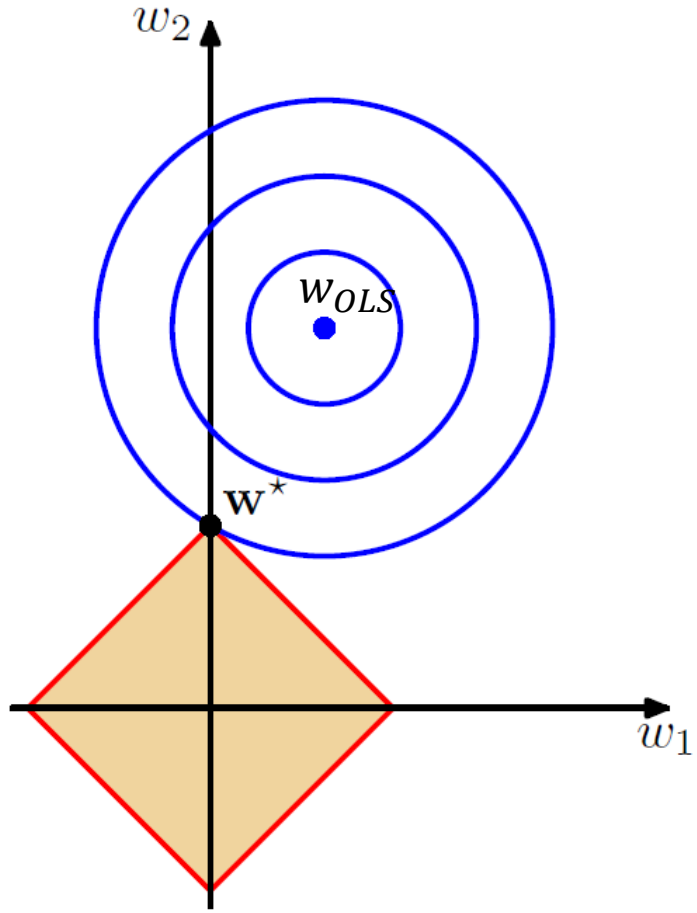
Diameter of circular contour depends on λ

LASSO (L1 Regularisation)

- Least Absolute Shrinkage and Selection Operator
- LASSO regression objective:

$$\min_{\boldsymbol{w}} (\boldsymbol{y} - \boldsymbol{X}^*\boldsymbol{w})^T (\boldsymbol{y} - \boldsymbol{X}^*\boldsymbol{w}) + \lambda ||\boldsymbol{w}||_1$$

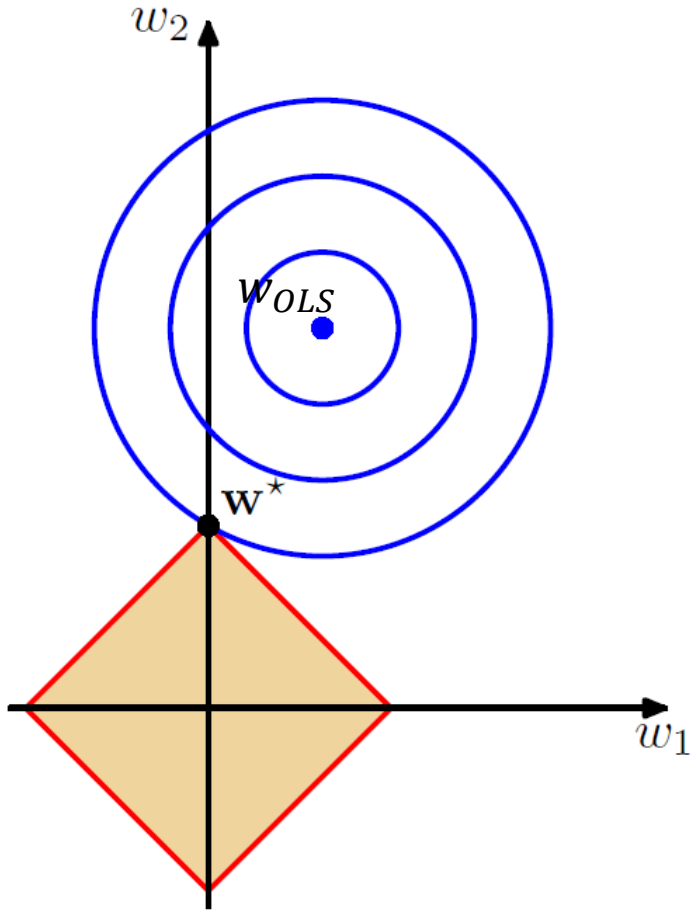
- Can drive some coefficients exactly to zero
- Can be used for feature selection
- Geometric View:
  - OLS: Minimises error in an unconstrained space
  - Ridge: Minimises error constrained inside a L1 square



LASSO Solution using Coordinate Descent

- Issue: LASSO objective is not differentiable at zero and has no closed form solution
- Idea of Coordinate Descent:
  - Optimize the objective one parameter at a time while holding the others fixed

- Algorithm Steps:
  1. Standardise each input feature
  2. Initialise with random small values (or zeros) of parameters $\boldsymbol{w}^0$



LASSO using Coordinate Descent

- Algorithm Steps:
  3. Repeat until convergence:
     a) For each of the D predictors (inputs), compute partial residual for all samples

$$r_{i(-j)} = y_i - \sum_{k \neq j} x_{ik} w_k \;\; \forall \;\; j \; vars$$

        - Gives what part of y is left unexplained if we ignore variable j
        - This is the target w_j must explain

     b) Compute OLS like contribution of feature j
        - Regress this residual only on $\boldsymbol{x}_j$ (all samples)

$$\boldsymbol{r}_{-j} = \rho_j \boldsymbol{x}_j$$

$$\rho_j = \frac{\boldsymbol{x}^T{}_j \boldsymbol{r}_{-j}}{\boldsymbol{x}_j^T \boldsymbol{x}_j} = \boldsymbol{x}^T{}_j \boldsymbol{r}_{-j} \text{ (scaled features)}$$

        - Explains potential value for w_j when there is no L1 penalty

LASSO using Coordinate Descent

- Algorithm Steps:
  3. Repeat until convergence:
     c) Apply Soft Thresholding and update the weights as follows:

$$w_j = \frac{\rho_j - \lambda}{\left|\left|x_j\right|\right|^2} \; ; \rho_j > \lambda$$

$$w_j = 0 \; ; |\rho_j| \leq \lambda$$

$$w_j = \frac{\rho_j + \lambda}{\left|\left|x_j\right|\right|^2} \; ; \rho_j < -\lambda$$

  - No gradients, no matrix inverse and natural sparsity
  - Note: λ is chosen using validation techniques

Model Validation

Model Validation

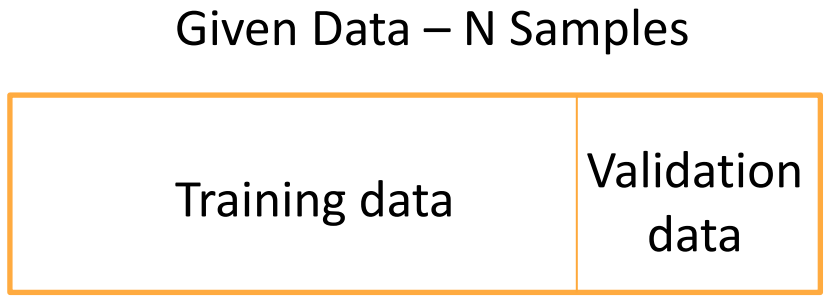
- Hyperparameters are to be chosen by the user
  - Regularisation rate
  - Regularisation type: Ridge or LASSO
  - Learning rate
  - Polynomial degree
- Alternately, it is chosen using a model validation strategy
- To perform model validation and testing, the given dataset is split into three subsets:
  - Training set: Set of samples used to learn the model parameters. So loss function is minimised over this dataset
  - Validation set: Set of samples used for model comparison and selection
  - Test set: Set of samples used to compute unbiased performance of the selected model

Model Validation Techniques

- Different model validation techniques exist which can be used to chosen a good model:
  - Hold out
  - K-fold cross validation
  - Leave-one-out cross validation

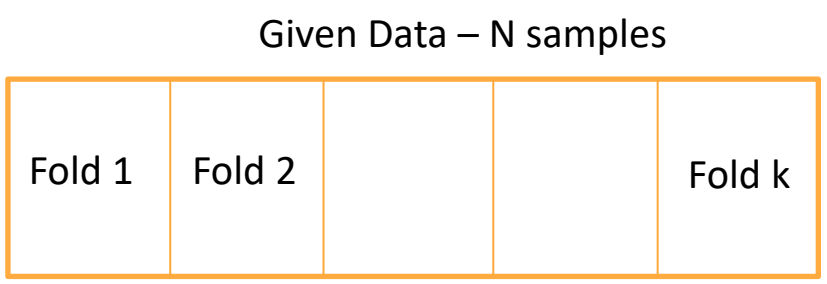
- All validation techniques involve splitting the given data into training and validation sets and validating using one of the metrics

Model Validation: Hold out

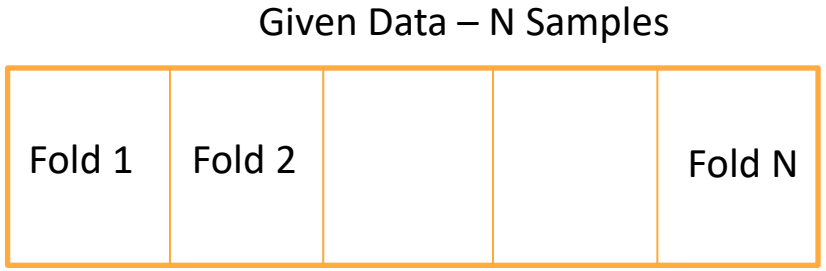


- Most basic validation method where given data is split into training and validation sets in the ratio of 80:20 or 70:30
- Issue: Randomly splitting the data may result in sampling bias
- The distribution of training and validation data sets may not represent the same population distribution from the original data is drawn

Model Validation: K-fold and Leave-one-out



- If there are N samples in the dataset, then they are split into k folds each with $\frac{N}{k}$ samples
- Each time, one fold is kept out of the training set and is used to validate the model after training
- Repeated by considering each of the folds as validation sets and average performance is taken
- This reduces sampling bias to considerable extent



- A special case of k fold cross validation in which $k = N$
- Used when number of samples are very less in number

Metrics for Regression Model Evaluation

Metrics to Measure Prediction Error

- Mean Absolute Error (MAE): Mean of the absolute values of residuals

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$$

  - Easy to understand and same units as dependent variable
  - Treats all errors equally and does not amplify large errors or outliers
- Root Mean Squared Error (RMSE): Square root of the mean of squared residuals

$$RMSE = \sqrt{\left(\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2\right)}$$

  - Penalizes large errors and useful to detect outliers or when large errors are critical
  - Same unit as output

Metrics to Measure Relative Error

- Mean Absolute Percentage Error (MAPE): Error as percentage of actual value of dependent variable

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100$$

  - Useful for understanding relative errors and easy to interpret since it is percentage
  - Can be very high when actual values are small - to be avoided
  - Undefined when $y_i = 0$
  - Scale independent and can be used to compare models and outputs at different scales

Metrics to Evaluate Goodness of Fit

- R-squared (R^2): Metric which indicates how good is the fit

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}$$

  - TSS: Represents total variation in the dependent variable relative to its mean
  - RSS: Represents the unexplained variation in the dependent variable by the model
  - Indicates proportion of variance in the output that is explained by the model
  - R^2 of 1 indicates perfect fit where there are no residuals
  - Negative R^2 indicates worse fit where mean prediction is better than model prediction
  - Note: To be used only to understand variance explained and compare models on same dataset
  - Issues:
    - Can be misleading in case of low variance in the dependent variable
    - R^2 increases with additional variables even if irrelevant
    - High R^2 does **not imply good predictions** (model may still have large errors)

Feature Selection Methods

Feature Selection

- Objective: Select a subset of the features that minimises generalisation or validation error

$$S \subset \{1,2,\dots,D\}$$

$$\mathcal{L}_{val}(S) = \frac{1}{N_{val}} \sum_i \left(y_i - \hat{y}_i^{(s)}\right)^2$$

- Issue: Search over all possible subsets is not feasible
- Approximate approaches:
  - Recursive Feature Elimination: Start with all features and remove least important steps in a recursive fashion
  - Sequential Feature Selection: Build subset step by step based on validation loss
- Computationally expensive but generally better than LASSO

Recursive Feature Elimination

- Idea: Important of feature j is proportional to the associated coefficient w_j
- Small $|w_j| \rightarrow$ Weak contribution $\rightarrow$ Candidate for removal
- Algorithm:
  - Let $S_0 = \{1,2, \dots, D\}$
  - For each iteration $i = 0,1,2, \dots$
    - Fit Linear model using all features in S_i
    - Rank the features based on their coefficients: $r_j = |w_j^{(i)}|$
    - Remove feature with smallest rank
    - Repeat until desired number of features remain

- Greedy backward elimination as an approximation to best subset selection
- Works well when features are not highly correlated

Sequential Feature Selection

- Here, the validation loss is directly minimized

$$\mathcal{L}_{val}(S) = \frac{1}{N_{val}} \sum_{i} \left(y_i - \hat{y}_i^{(s)} \right)^2$$

- Algorithm:
  - Start with an empty feature set $S_0 = \Phi$
  - At step i:
    - For each candidate feature j not in S_i
    - Fit model using $S_i \cup \{j\}$
    - Compute Validation loss $\mathcal{L}_{val}(S_i \cup \{j\})$
    - Choose feature: $j^* = \underset{j}{\operatorname{argmin}}\, \mathcal{L}_{val}(S_i \cup \{j\})$
    - Update the feature set: $S_{i+1} = S_i \cup \{j^*\}$
- Note: This can also be done in a backward manner by starting with complete set of features
- Use when validation performance is the primary goal and when features are correlated

Bayesian Linear Regression

Bayesian Linear Regression: Key Idea

- Linear Model

$$y = w_0 + w_1\phi_1(\boldsymbol{x}) + w_2\phi_2(\boldsymbol{x}) + \cdots + w_D\phi_D(\boldsymbol{x}) + \epsilon$$

- Assumption on noise: $\epsilon \backsim \mathcal{N}(0, \beta^{-1})$
- Implies target values follow a Gaussian distribution

$$p(\boldsymbol{y}|\boldsymbol{w}) = \mathcal{N}(\boldsymbol{X}\boldsymbol{w}, \beta^{-1}\boldsymbol{I})$$

- β: precision or inverse variance
- Bayesian Idea:
  - Estimate a distribution over parameters instead of point estimates
  - Assumes a prior distribution on parameters and infers a posterior based on data using Baye's rule

$$p(\boldsymbol{w}|\mathcal{D}) = \frac{p(\mathcal{D}|\boldsymbol{w})p(\boldsymbol{w})}{p(\mathcal{D})}$$

Prior and Posterior

- Prior Assumption:

$$p(\boldsymbol{w}) = \mathcal{N}(0, \alpha^{-1}\boldsymbol{I})$$

- Other Assumptions:
  - Inputs are treated as fixed and certain
  - Only the outputs are modelled probabilistically
  - For simplicity $\mathcal{D}$ is replaced with $\boldsymbol{y}$ in the probabilities

- Since prior and likelihood are gaussian, posterior is also a gaussian

$$p(\boldsymbol{w}|\boldsymbol{y}) = \mathcal{N}(\boldsymbol{m}_N, \boldsymbol{S}_N)$$

- Instead of a single weight vector, we have a distribution over weights

Posterior Mean and Covariance

- Posterior distribution:

$$p(\boldsymbol{w}|\boldsymbol{y}) = \mathcal{N}(\boldsymbol{m}_N, \boldsymbol{S}_N)$$

- Posterior Gaussian's Covariance matrix:

$$\boldsymbol{S}_N = \left(\alpha \boldsymbol{I} + \beta \boldsymbol{X}^T \boldsymbol{X}\right)^{-1}$$

- Interpretation: Posterior precision = Prior precision + Data precision
- Posterior Gaussian's mean vector:

$$\boldsymbol{m}_N = \beta \boldsymbol{S}_N \boldsymbol{X}^T y$$

$$\boldsymbol{m}_N = \left(\alpha \boldsymbol{I} + \beta \boldsymbol{X}^T \boldsymbol{X}\right)^{-1} \beta \boldsymbol{X}^T y$$

- Interpretation: Equivalent to ridge regression solution with $\lambda = \frac{\alpha}{\beta}$
- In other words, large prior precision or large noise variance acts as strong regularisation
- Note: Ridge regression can be interpreted as MAP estimation

Predictions

- Consider a new sample $(\boldsymbol{x}_{new}, y_{new})$
- From model assumption:

$$p(y_{new}|\boldsymbol{w}) = \mathcal{N}(\boldsymbol{x}_{\boldsymbol{new}}^{\boldsymbol{T}}\boldsymbol{w}, \beta^{-1})$$

- Above describes the prediction for a fixed $\boldsymbol{w}$

- Since $\boldsymbol{w}$ is uncertain with known distribution, prediction is averaged over the posterior

$$p(y_{new}|x_{new}, \mathcal{D}) = \int p(y_{new}|\boldsymbol{w})p(\boldsymbol{w}|\mathcal{D})dw$$

$$p(y_{new}|x_{new}, \mathcal{D}) = \mathcal{N}(\boldsymbol{x}_{\boldsymbol{new}}^{\boldsymbol{T}}\boldsymbol{m}_N, \beta^{-1} + \boldsymbol{x}_{new}^{T}\boldsymbol{S}_N\boldsymbol{x}_{new})$$

Non-Linear Regression

Non-Linear Regression

- Suppose there exists a non-linear regression model which structure is as follows:

$$\ln y = A - \frac{B}{x+C} \text{ : Antoine Eq.}$$

- A,B,C are the parameters and x, y are the variables
- Objective is to estimate A, B, C given the data (n samples) pertaining to x, y
- Cannot be converted to a linear form in terms of the parameters

Non-Linear Least Squares Solution

- Objective: Minimise the sum of residuals between observed and predicted values

$$\min_{A,B,C} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

$$s.t. \ln \hat{y}_i = A - \frac{B}{x_i + C}$$

- Steps:
  - Choose initial values for A, B, C
  - Use a non-linear least squares solver (Gradient descent or Gauss-Newton) to minimise the objective and obtain A, B, C
- Note: scipy library in python has this solver inbuilt

Topics for Further Reading in Regression

- Generalised Least Squares
- Elastic Net : L1 + L2 regularisation of a linear regression model
- Confidence intervals for parameters and predictions